# Chapter 5.8 - Memory Hierarchy Framework & Block Placement

Comprehensive Revision Flashcards • 90 Key Concepts & Practice Questions

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| #1 | **Q1: What are the four fundamental questions that provide a common framework for understanding any memory hierarchy?** |
| | **Answer:** 1. Where can a block be placed? 2. How is a block found? 3. Which block should be replaced on a miss? 4. How are writes handled? |

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| #2 | **Q2: In a memory hierarchy, what is the term for a block placement scheme where a block can be placed in exactly one location?** |
| | **Answer:** Direct mapped. |

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| #3 | **Q3: In a _____ placement scheme, a block can be placed in a restricted set of places in the cache.** |
| | **Answer:** set-associative |

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| #4 | **Q4: What placement scheme allows a block to be placed anywhere in the upper level of the memory hierarchy?** |
| | **Answer:** Fully associative. |

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| #5 | **Q5: What is the primary advantage of increasing the degree of associativity in a memory hierarchy?** |
| | **Answer:** It usually decreases the miss rate by reducing misses that compete for the same location. |

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| #6 | **Q6: According to the source, what is the typical percentage reduction in miss rate when moving from a direct-mapped to a two-way set-associative cache?** |
| | **Answer:** Between 20% and 30%. |

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| #7 | **Q7: What are the two potential disadvantages of increasing associativity in a cache?** |
| | **Answer:** Increased cost and slower access time. |

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| #8 | **Q8: Why does the absolute improvement in miss rate from associativity shrink as cache sizes increase?** |
| | **Answer:** Because the overall miss rate of a larger cache is already lower, leaving less opportunity for improvement. |

| #9 | **Q9: How is a block found in a direct-mapped cache?** |
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| | **Answer:** By using an index, which requires only one comparison. |

| #10 | **Q10: How is a block located in a set-associative cache?** |
|---|---|
| | **Answer:** By first indexing the set and then searching among the elements within that set. |

| #11 | **Q11: What type of search is required to find a block in a fully associative cache?** |
|---|---|
| | **Answer:** A full search of all cache entries. |

| #12 | **Q12: What data structure is used to find a block in a virtual memory system?** |
|---|---|
| | **Answer:** A separate lookup table, specifically the page table. |

| #13 | **Q13: Why are fully associative caches typically prohibitive except for small sizes?** |
|---|---|
| | **Answer:** The cost of the comparators required for a full search becomes overwhelming for large caches. |

| #14 | **Q14: What block placement scheme is almost always used for virtual memory systems?** |
|---|---|
| | **Answer:** Fully associative placement. |

| #15 | **Q15: What are the three motivating factors for using fully associative placement in virtual memory systems?** |
|---|---|
| | **Answer:** Misses are very expensive, it allows for sophisticated software replacement schemes, and the map can be easily indexed. |

| #16 | **Q16: What is the primary advantage of a direct-mapped cache design?** |
|---|---|
| | **Answer:** Faster access time and simplicity, as finding the block does not depend on a comparison. |

| #17 | **Q17: In an associative cache, which replacement strategy involves selecting a candidate block at random?** |
|---|---|
| | **Answer:** Random replacement. |

| #18 | **Q18: In an associative cache, what does the Least Recently Used (LRU) replacement strategy dictate?** |
|---|---|
| | **Answer:** The block that has been unused for the longest time should be replaced. |

| #19 | **Q19: Why is a true LRU replacement policy costly to implement for caches with high associativity (more than four-way)?** |
|---|---|
| | **Answer:** Because tracking the detailed usage information for many blocks is expensive in hardware. |

| #20 | **Q20: For a two-way set-associative cache, how does the miss rate of random replacement compare to LRU replacement?** |
|---|---|
| | **Answer:** The miss rate for random replacement is about 1.1 times higher than for LRU. |

| #21 | **Q21: What type of replacement algorithm is always approximated in virtual memory, and why?** |
|---|---|
| | **Answer:** Some form of LRU is always approximated because the cost of a miss is enormous, making even a tiny reduction in miss rate important. |

| #22 | **Q22: What is a 'write-through' policy in a memory hierarchy?** |
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| | **Answer:** Information is written to both the block in the cache and the block in the lower level of the memory hierarchy simultaneously. |

| #23 | **Q23: What is a 'write-back' policy in a memory hierarchy?** |
|---|---|
| | **Answer:** Information is written only to the block in the cache, and the modified block is written to the lower level only when it is replaced. |

| #24 | **Q24: What are the three key advantages of a write-back policy?** |
|---|---|
| | **Answer:** Processor can write at cache speed, multiple writes to a block require only one lower-level write, and it allows for high-bandwidth transfers. |

| #25 | **Q25: What are the two main advantages of a write-through policy?** |
|---|---|
| | **Answer:** Misses are simpler and cheaper, and the policy is easier to implement. |

| #26 | **Q26: Why is a write-back policy considered the only practical option for virtual memory systems?** |
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| | **Answer:** Because of the very long latency of a write to the lower level of the hierarchy (e.g., a disk). |

| #27 | **Q27: Which write policy is typically used by lowest-level caches today, and why?** |
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| | **Answer:** Write-back, because the rate of processor writes generally exceeds the rate at which the memory system can process them. |

| #28 | **Q28: What is the 'three Cs' model used for?** |
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| | **Answer:** It provides an intuitive model for classifying cache misses into one of three categories to understand memory hierarchy behavior. |

| #29 | **Q29: What are the three categories of misses in the 'three Cs' model?** |
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| | **Answer:** Compulsory misses, capacity misses, and conflict misses. |

| #30 | **Q30: Define: Compulsory Miss** |
|---|---|
| | **Answer:** A cache miss caused by the first access to a block that has never been in the cache; also called a cold-start miss. |

| #31 | **Q31: Define: Capacity Miss** |
|---|---|
| | **Answer:** A cache miss that occurs because the cache cannot contain all the blocks needed during a program's execution, even with full associativity. |

| #32 | **Q32: Define: Conflict Miss** |
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| | **Answer:** A miss in a direct-mapped or set-associative cache that occurs when multiple blocks compete for the same set and would be eliminated in a fully associative cache of the same size. |

| #33 | **Q33: What is the primary way to reduce the number of compulsory misses?** |
|---|---|
| | **Answer:** Increase the block size. |

| #34 | **Q34: What is a straightforward way to reduce capacity misses?** |
|---|---|
| | **Answer:** Enlarge the cache size. |

| #35 | **Q35: How can conflict misses be reduced in a cache design?** |
|---|---|
| | **Answer:** By increasing the associativity. |

| #36 | **Q36: What is the potential negative performance effect of increasing cache size to reduce capacity misses?** |
|---|---|
| | **Answer:** It may increase the cache access time. |

| #37 | **Q37: What is the potential negative performance effect of increasing associativity to reduce conflict misses?** |
|---|---|
| | **Answer:** It may increase the cache access time. |

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| #38 | **Q38: What are the potential negative performance effects of increasing the block size to reduce compulsory misses?** |
| | **Answer:** It increases the miss penalty, and a very large block size could increase the overall miss rate. |

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| #39 | **Q39: According to the 'Check Yourself' section, is it true that there is no way to reduce compulsory misses?** |
| | **Answer:** No, this is false; increasing the block size can reduce compulsory misses. |

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| #40 | **Q40: According to the 'Check Yourself' section, is it true that fully associative caches have no conflict misses?** |
| | **Answer:** Yes, this is true by definition, as conflict misses are those eliminated by a fully associative cache. |

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| #41 | **Q41: As cache sizes grow, the relative improvement from associativity _____, and the absolute improvement _____** |
| | **Answer:** increases only slightly; shrinks significantly |

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| #42 | **Q42: Why do smaller caches obtain a significantly larger absolute benefit from associativity?** |
| | **Answer:** Because the base miss rate of a small cache is larger, providing more opportunity for improvement. |

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| #43 | **Q43: The method of finding a block in a memory hierarchy is dictated by the _____ scheme.** |
| | **Answer:** block placement |

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| #44 | **Q44: What feature of on-chip L2 caches enables them to have much higher associativity compared to off-chip caches?** |
| | **Answer:** Their hit times are not as critical, and designers are not constrained by standard SRAM chip building blocks. |

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| #45 | **Q45: In a direct-mapped cache, how many candidate blocks are there for replacement on a miss?** |
| | **Answer:** Only one. |

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| #46 | **Q46: In virtual memory systems, what hardware feature is often provided to help the OS approximate an LRU replacement policy?** |
| | **Answer:** Reference bits or equivalent functionality to track page usage. |

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| #47 | **Q47: In the 'three Cs' model, the total miss rate of a direct-mapped cache is the sum of compulsory, capacity, and conflict misses from one-way, two-way, and _____ associativity.** |
| | **Answer:** four-way |

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| #48 | **Q48: What are the four fundamental questions that provide a common framework for understanding any level of a memory hierarchy?** |
| | **Answer:** 1. Where can a block be placed? 2. How is a block found? 3. Which block should be replaced on a miss? 4. What happens on a write? |

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| #49 | **Q49: In a direct-mapped placement scheme, how many sets are there and how many blocks are in each set?** |
| | **Answer:** There are as many sets as blocks in the cache, with exactly one block per set. |

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| #50 | **Q50: In a fully associative placement scheme, how many sets are there and how many blocks are in each set?** |
| | **Answer:** There is only one set, which contains all the blocks in the cache. |

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| #51 | **Q51: What is the primary advantage of increasing the degree of associativity in a cache?** |
| | **Answer:** It usually decreases the miss rate by reducing misses that compete for the same location (conflict misses). |

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| #52 | **Q52: What is the typical percentage reduction in miss rate when moving from a direct-mapped (one-way) to a two-way set-associative cache?** |
| | **Answer:** The miss rate is typically reduced by 20% to 30%. |

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| #53 | **Q53: What are the two potential disadvantages of increasing a cache's associativity?** |
| | **Answer:** Increased cost and slower access time. |

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| #54 | **Q54: As cache sizes grow, the _____ improvement in the miss rate from associativity shrinks significantly.** |
| | **Answer:** absolute |

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| #55 | **Q55: How is a block found in a set-associative cache?** |
| | **Answer:** By indexing to find the correct set and then searching among the elements within that set. |

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| #56 | **Q56: How is a block found in a fully associative cache?** |
| | **Answer:** By searching through all entries in the entire cache. |

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| #57 | **Q57: What mechanism is used to find a block (page) in virtual memory systems?** |
| | **Answer:** A separate lookup table, known as the page table. |

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| #58 | **Q58: What is the first reason virtual memory systems almost always use fully associative placement?** |
| | **Answer:** Full associativity is beneficial because memory misses are extremely expensive in terms of performance. |

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| #59 | **Q59: What is the second reason virtual memory systems use fully associative placement?** |
| | **Answer:** It allows software to use sophisticated replacement schemes to reduce the miss rate. |

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| #60 | **Q60: What is the third reason virtual memory systems use fully associative placement?** |
| | **Answer:** The full map (page table) can be easily indexed with no extra hardware or searching required. |

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| #61 | **Q61: What is the primary advantage of a direct-mapped cache that might lead a designer to choose it?** |
| | **Answer:** Its advantage in access time and simplicity, as finding a block does not depend on a comparison. |

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| #62 | **Q62: For which types of caches must a block replacement policy be decided upon when a miss occurs?** |
| | **Answer:** Set-associative and fully associative caches. |

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| #63 | **Q63: What are the two primary strategies for block replacement in associative caches?** |
| | **Answer:** Random and Least Recently Used (LRU). |

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| #64 | **Q64: Define the Least Recently Used (LRU) replacement policy.** |
| | **Answer:** The block that is replaced is the one that has been unused for the longest time. |

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| #65 | **Q65: Why is a true LRU policy too costly to implement for caches with a high degree of associativity (more than four-way)?** |
| | **Answer:** Because tracking the detailed usage information required for true LRU is expensive in terms of hardware. |

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| #66 | **Q66: For a two-way set-associative cache, how does the miss rate of a random replacement policy compare to an LRU policy?** |
| | **Answer:** The miss rate for random replacement is about 1.1 times higher than for LRU replacement. |

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| #67 | **Q67: Why is some form of LRU always approximated for replacement in virtual memory?** |
| | **Answer:** Because the cost of a miss is enormous, so even a tiny reduction in the miss rate is very important. |

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| #68 | **Q68: The _____ write policy involves writing information to both the block in the cache and the block in the lower level of the memory hierarchy.** |
| | **Answer:** write-through |

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| #69 | **Q69: The _____ write policy involves writing information only to the block in the cache, and to the lower level only when that block is replaced.** |
| | **Answer:** write-back |

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| #70 | **Q70: What is an advantage of write-back concerning processor write speed?** |
| | **Answer:** Individual words can be written at the speed of the cache, not the slower main memory. |

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| #71 | **Q71: What is an advantage of write-back concerning multiple writes to the same block?** |
| | **Answer:** Multiple writes within a single block require only one write to the lower-level memory when the block is finally replaced. |

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| #72 | **Q72: What is an advantage of write-through concerning cache misses?** |
| | **Answer:** Misses are simpler and cheaper because they never require a modified block to be written back to the lower level. |

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| #73 | **Q73: Why is write-back the only practical write policy for virtual memory systems?** |
| | **Answer:** Because of the extremely long latency of a write to the lower level of the hierarchy (e.g., a disk). |

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| #74 | **Q74: Which write policy do lowest-level caches typically use today and why?** |
| | **Answer:** Write-back, because the rate of processor writes generally exceeds the rate at which the memory system can process them. |

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| #75 | **Q75: What are the "Three Cs" used to classify all cache misses?** |
| | **Answer:** Compulsory misses, capacity misses, and conflict misses. |

| #76 | **Q76: Term: Compulsory miss** |
|---|---|
| | **Answer:** A cache miss caused by the first access to a block that has never been in the cache; also called a cold-start miss. |

| #77 | **Q77: Term: Capacity miss** |
|---|---|
| | **Answer:** A cache miss that occurs because the cache cannot contain all the blocks needed by the program, even with full associativity. |

| #78 | **Q78: Term: Conflict miss** |
|---|---|
| | **Answer:** A cache miss in a direct-mapped or set-associative cache that occurs when multiple blocks compete for the same set. |

| #79 | **Q79: A conflict miss would be eliminated if the cache were _____ of the same size.** |
|---|---|
| | **Answer:** fully associative |

| #80 | **Q80: What is the most direct way to reduce the number of conflict misses?** |
|---|---|
| | **Answer:** By increasing the associativity of the cache. |

| #81 | **Q81: What is the most direct way to reduce the number of capacity misses?** |
|---|---|
| | **Answer:** By enlarging the total size of the cache. |

| #82 | **Q82: What is a potential negative performance impact of increasing the cache size?** |
|---|---|
| | **Answer:** It may increase the cache access time. |

| #83 | **Q83: What is a potential negative performance impact of increasing the block size?** |
|---|---|
| | **Answer:** It increases the miss penalty (the time to fetch the larger block). |

| #84 | **Q84: According to the 'Check Yourself' section, do fully associative caches have conflict misses?** |
|---|---|
| | **Answer:** No, by definition, fully associative caches have no conflict misses. |

| #85 | **Q85: As cache associativity increases from four-way to eight-way, what happens to the improvement in miss rate?** |
|---|---|
| | **Answer:** There is very little improvement in the miss rate compared to the gain from one-way to two-way. |

| #86 | **Q86: What is a key trade-off that makes memory hierarchy design challenging?** |
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| | **Answer:** Every change that can improve the miss rate (like increasing size or associativity) can also negatively affect overall performance (like increasing access time). |

| #87 | **Q87: In the "Three Cs" model graph (Figure 5.36), which component of the miss rate is so small it is often not visible?** |
|---|---|
| | **Answer:** The compulsory miss component. |

| #88 | **Q88: Why have first-level caches been growing slowly, if at all, compared to second-level caches?** |
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| | **Answer:** To avoid increasing access time, which could lower overall performance, as first-level cache access time is often critical to processor cycle time. |

| #89 | **Q89: In a set-associative cache, how many blocks are candidates for replacement on a miss?** |
|---|---|
| | **Answer:** All the blocks within the specific set where the miss occurred. |

| #90 | **Q90: How is the location method for a block in a memory hierarchy determined?** |
|---|---|
| | **Answer:** It is dictated by the block placement scheme (direct-mapped, set-associative, or fully associative). |